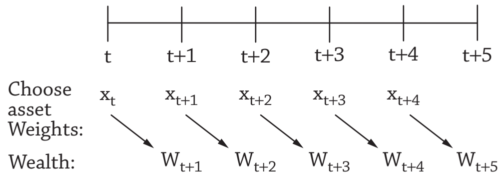

Figure 4.1

potentially change due to pre-determined variables, like investor constraints or liabilities changing, or due to time-varying investment returns, like booms versus busts.

The investor wishes to maximize the expected utility of end of period wealth at time T by choosing a dynamic series of portfolio weights:

$$\max_{\{x_t\}} E[U(W_T)], \tag{4.2}$$

subject to constraints. Some examples of constraints are that an investor may not be able to short (this is a *positivity constraint* so $x_t \geq 0$), may not be able to lever (so the portfolio weight is bounded, $0 \leq x_t \leq 1$), or can only sell a certain portion of her portfolio each period (this is a *turnover constraint*). Although the portfolio weights $x_{t+\tau}$ are, of course, only implemented at time $t + \tau$, the complete set of weights $\{x_t\}$ from t to T–1 is chosen at time t, the beginning of the problem. The set of optimal weights can be quite complicated: they may not only vary through time as the horizon approaches, but they may vary by state. For example, x_t could take on two values at time t: hold 50% in equities if we are in a recession and 70% if we are in a bull market. The complete menu of portfolio strategies across time and states is determined at the start. Thus, the optimal dynamic trading strategy is completely known from the beginning, even though it changes through time: as asset returns change, the strategy optimally responds, and as utility and liabilities change, the strategy optimally responds.

For the remainder of this chapter, we work with constant relative risk aversion (CRRA) utility (see chapter 2):

$$E[U(W)] = E\left[\frac{W^{1-\gamma}}{1-\gamma}\right], \tag{4.3}$$

where W is the investor's wealth at the end of the period and γ is her risk aversion coefficient. In what follows, I will omit the $1 - \gamma$ term in the denominator of equation (4.3) because maximizing $E[W^{1-\gamma}/(1-\gamma)]$ is exactly the same as maximizing $E[W^{1-\gamma}]$. CRRA is locally mean-variance so the risk aversion γ has the same meaning in mean-variance utility, U^{MV} (see chapter 3):

$$U^{MV} = E(r_p) - \frac{\gamma}{2}\mathrm{var}(r_p), \tag{4.4}$$